# Evan L. Sticca

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Bioinformatics Expert Specializing in Clinical Translational Science and Data-Driven Solutions

## Education

**University of Colorado Anschutz Medical Campus** Aug 2018 – Jul 2023

PhD, Human Medical Genetics and Genomics Program **-** Mentored by Dr. Christopher Gignoux

- Achieved Accredited Graduate Certificate in Research Compliance and Project Management – Dec 2021
- Completed Clinical Data Science Coursera Specialization – Jan 2022
- Representative for Graduate Student Advisory Group to the Dean, Graduate Student Council, Student Advocates for Mental Health, Movember Foundation Ambassador, PhD Post newspaper
- Relevant Courses: Biostatistical Methods I/II, Responsible Conduct of Research, Human Genetics, Practical Biological Data Analysis, Legal and Regulatory Environment of Life Science Innovations, Biomedical Entrepreneurship

**State University of New York at Stony Brook** Aug 2013 – Dec 2014

Master’s Degree, Department of Ecology and Evolution, Concentration in Applied Evolution

- Graduate Student Researcher - Mentored by Dr. Brenna Henn
- The Gotham Seminar on Genomics and Statistics, Stony Brook GSO Member, Ballroom Dancing Club
- Relevant Courses: Applied Ecology and Evolution, Ecological Genetics and Genomics, Biometry, Evolutionary Ecology, Graduate Genetics, Computational Biology

**University of Colorado at Boulder** Aug 2009 – Aug 2012

Bachelor’s Degree in Evolutionary Biology and Ecology

- Arnett Honors Housing, CU Program Council, CU Fencing Club
- Arts and Sciences Dean’s List Spring 2012

## Employment

**Castle Biosciences: Data Architect** - (July 2023 – Present)

I am the principal informatics developer on a digital pathology advanced laboratory test. I am responsible for initiating pipeline improvements and I prompt requirements gathering for further input from stakeholders. Overall, I improve output of clinically relevant results by maximizing efficiency in data flows, algorithmic performance and computational architecture. Some specific projects I am proud of include reformatting workflows to use raw data inputs, thus reducing cost of goods upwards of $100,000 annually. I further reduced redundant inputs by a minimum of 25% thus improving efficiency. I re-developed quality control tools to work on scalable computational architecture, and re-engineering laboratory information management systems to avoid redundancy and reduced the potential for mismatching results in clinical reports. One final project worth highlighting is identifying expected distributions for all test bio marker values and developing acceptance criteria to alert the laboratory operations team when any deviation from the distribution occurs. This lead to identifying of a critical laboratory deficiency, to which I proposed a potential in silico solution, and demonstrated proof of concept via simulation experiments. I then fast-tracked algorithm re-development and completed it 2 months ahead of schedule, helping expedite further improvements and enterprise decision making. I also applied these changes retrospectively as well to improve all previously issued clinical reports, providing higher quality data to primary care physicians.

**University of Colorado Anschutz: Research Assistant** - (August 2018-July 2023)

In addition to full time PhD education, I also served as a full time Research Assistant under the UCHealth BioBank. My responsibilities included delivering phenotype and covariate data from electronic health records to researchers requesting data for genetic association studies. I used Google Analytics BigQuery and SQL frameworks to extract data from Google S3 buckets, then performed data quality control, cleaning and preparation. I delivered high quality phenotype reports to scientists,. Contingent upon access to the data, I also directed users to appropriate training modules and ensured they completed their courses before granting access. I further collaborated with them for what statistical models to use, i.e. Least Squares Regression, Maximum Likelihood Estimation, etc. Finally, I supported grant writing in technical descriptions of methods and results to advance the funding and public data research accessibility for investigators looking to use our resources.

**Weill Cornell Institute for Precision Medicine: Bioinformatics Analyst** – (June 2016- June 2018)

Expanded the analytical platform in the Institute for Precision Medicine (IPM). Developed new software tools and built new pipelines for novel bioinformatics analyses in the context of clinically relevant results for cancer patients. Focused on expanding existing pipelines and production of novel ones regarding efficiency, robustness, reproducibility and interpretability. Prepared transcriptomics data from single cell RNA knock down experiments. Developed tools for visualization of results in an intuitive format. Finally, integrated diverse data to look for biologically relevant connections that might not otherwise be apparent. In all this work focused on the goal of using big data for precision medicine breakthroughs in patients' lives.

**The Rockefeller University: Lead Bioinformatics Specialist** – (March 2015 – June 2016, Promoted Aug 2015)

Developed bioinformatics pipelines for the Joseph Gleeson Laboratory of Pediatric Disease. Full job description available on request.

## Selected Research Publications (Google Scholar for full profile)

Host methylation predicts SARS-CoV-2 infection and clinical outcome -*Konigsberg et al. 2021, Nature Comm. Medicine*

EXaCT-2: Augmented Whole Exome Sequencing Optimized for Clinical Testing - *Hassane et al. 2019, Molecular Diagnostics*

Integrative Molecular Analysis of Patients With Advanced and Metastatic Cancer- *Sailer et al. 2019, JCO Precision Oncology*

Longitudinal Changes of One-Carbon Metabolites and Amino Acid Concentrations during Pregnancy in the Women First Maternal Nutrition Trial - *Gilley et al. 2019, Current Developments in Nutrition*

Autism risk in offspring can be assessed through quantification of male sperm mosaicism- *Breuss et al. 2019, Nature Medicine*

Integrating Clinical Genomics into Electronic Health Records to Foster Precision Medicine - *Sigaras et al. 2018, Molecular Diagnostics*

## Technical Skills (see LinkedIn for Certifications)

- Computing languages: R (13 years), Python (11 years), Bash scripting and Unix environment (11 years), MySQL (8 years), Tableau and Power BI (3 years), SAS, Java, C++, Hadoop, Databricks, Snowflake (working competency)
- Continuous Deployment of Pipelines to Cloud Computing environments (AWS, Microsoft Azure, GCP) via Docker
- Informatics tools: GATK, Platypus, TopHat, lumpy, Triodenovo, MuTect, Strelka, MIPgen, samtools, sambamba/samblaster, vcftools, bwa, XHMM, $H^3M^2$, PLINK/SEQ, snpEff/snpSift, MEGA, GREAT and Enrichr (Gene Ontology Analysis)
- Extensive familiarity in next generation sequencing technology, data processing and analysis pipelines
- Advanced competency in Spanish

## Additional Leadership and Experience

- Completed Agile Leadership Capstone November 2021
- Ambassador for Movember Foundation since November 2018
- Black Belt 1st dan of Go Ju Ryu since August 2009
- Eagle Scout of Boy Scouts of America since December 2005